321305

12425

3 Hours / 70 Marks Seat No. | | | | | | | | |

Instructions : (1) All Questions are *compulsory.*

(2) Figures to the right indicate full marks.

(3) Assume suitable data, if necessary.

(4) Use of Non-programmable Electronic Pocket Calculator is permissible.

Marks

1. Attempt any FIVE : **10**

(a) Name any four physical properties of pigments.

(b) Define a Pigment.

(c) State reason of blending pigments.

(d) Enlist any four chemical methods of pigment manufacturing.

(e) State in general sources of extenders.

(f) Give examples of reactive white pigments.

(g) List any four raw materials for black pigment manufacturing.

2. Attempt any THREE : **12**

(a) Differentiate between organic and inorganic pigments.

(b) Differentiate between dry and wet grinding technique.

(c) Write composition and properties of Blanc fixe extender.

(d) Explain titanium dioxide (A) not used for exterior application.

3. Attempt any THREE : **12**

(a) Classify pigments with examples of each category.

(b) Describe procedure of sampling pigments.

(c) Explain surface treatment for titanium oxide.

(d) Compare Channel Black and Furnace black.

4. Attempt any THREE : **12**

(a) Explain any four examples of pigment applications in ancient times.

(b) Explain the construction and working of Ball mill.

(c) Explain properties and applications of silicates.

(d) State properties and uses of zinc oxide pigment.

(e) Draw neat labelled diagram of Channel Black method for black pigment.

5. Attempt any TWO : **12**

(a) Explain in detail the procedure to determine Bulk Density of pigment as per IS-33.

(b) Describe formation of agglomerates and aggregates.

(c) Explain general flow sheet for extender pigment manufacturing.

6. Attempt any TWO : **12**

(a) Write chemical name, formulae, properties and application of China clay.

(b) Explain chloride process for Titanium di-oxide manufacturing with neat labelled flow sheet.

(c) Explain properties and applications of Black iron oxide.